

DeepSeek-V2

A Strong, Economical, and Efficient MoE Language Model

236B

Total Parameters

21B

Activated / Token

128K

Context Length

8.1T

Training Tokens

The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed

The Problem

Scaling model parameters improves intelligence — but at enormous cost.

Training Cost Explosion

Dense models require compute proportional to total parameters.
Training a 70B dense model demands massive GPU clusters for weeks.

KV Cache Bottleneck at Inference

Standard MHA stores full K/V pairs per head per token.
For long contexts (128K tokens), KV cache dominates GPU memory,
limiting batch size and throughput.

The Opportunity: MoE + Efficient Attention

Mixture-of-Experts activates a fraction of parameters per token,
decoupling model capacity from compute cost.
Latent attention compresses the KV cache without sacrificing quality.

DeepSeek-V2 Insight

Decouple capacity from cost

236B total parameters

for model capacity

21B activated per token

for compute efficiency

Compressed KV cache

for inference throughput

DeepSeek-V2 Architecture

MLA + DeepSeekMoE in a Transformer Block

Each Transformer block replaces:

Standard MHA → Multi-Head Latent Attention (MLA)

Low-rank KV compression into latent vectors

Decoupled RoPE for positional encoding

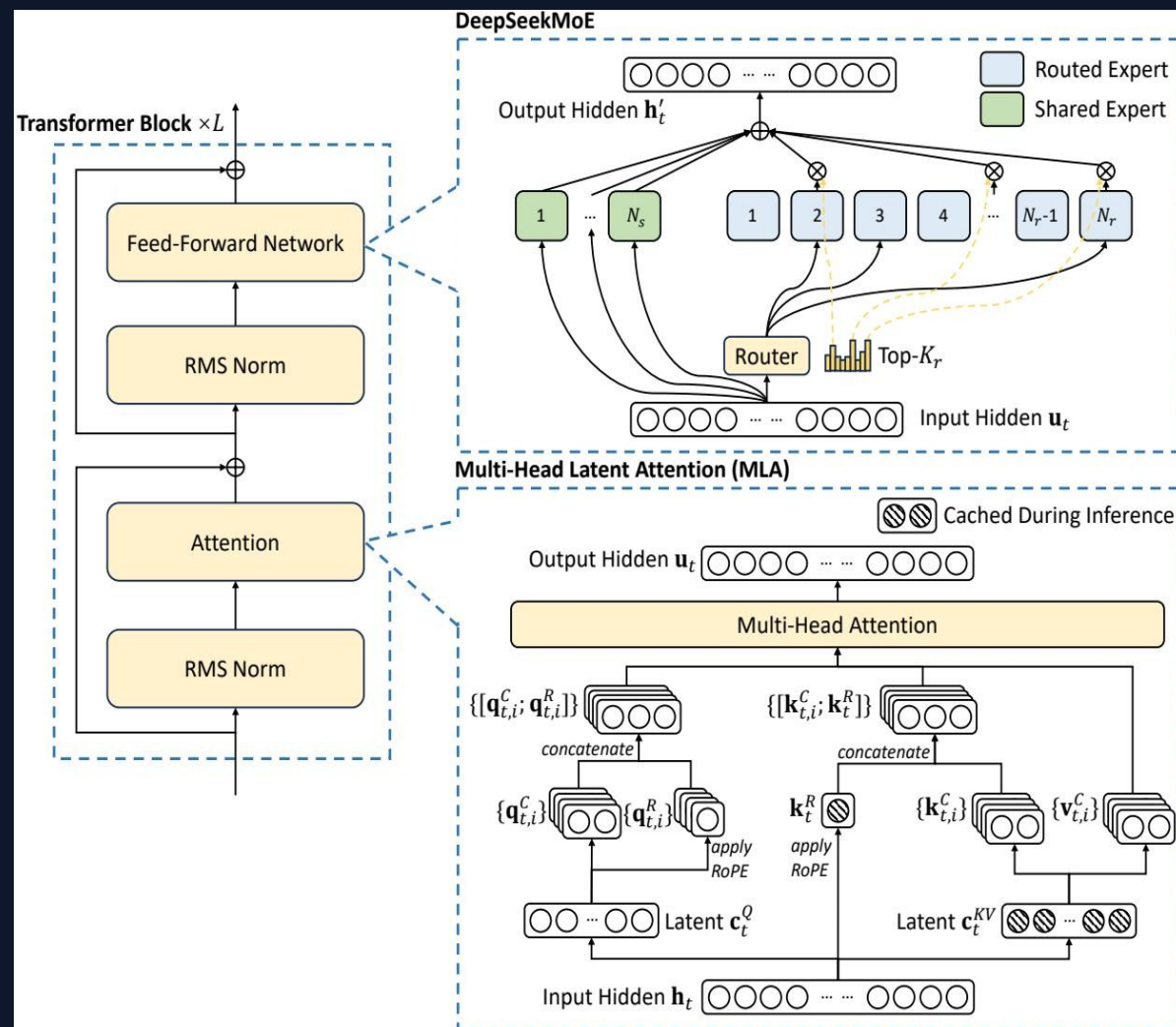
93.3% KV cache reduction vs. MHA

Dense FFN → DeepSeekMoE

Shared experts (always active) + Routed experts (sparse)

Fine-grained segmentation for specialization

Device-limited routing reduces communication



Multi-Head Latent Attention (MLA)

Compresses KV Cache by 93.3% While Outperforming MHA

Standard MHA Problem

Stores full K and V vectors per head per token.
Cache grows linearly with sequence length $\times$ heads.

MLA Solution: Low-Rank Joint Compression

Compress K and V into a compact latent vector c^{kv} .

Cache only the latent vector during inference.

Reconstruct full K, V on-the-fly via learned projections.

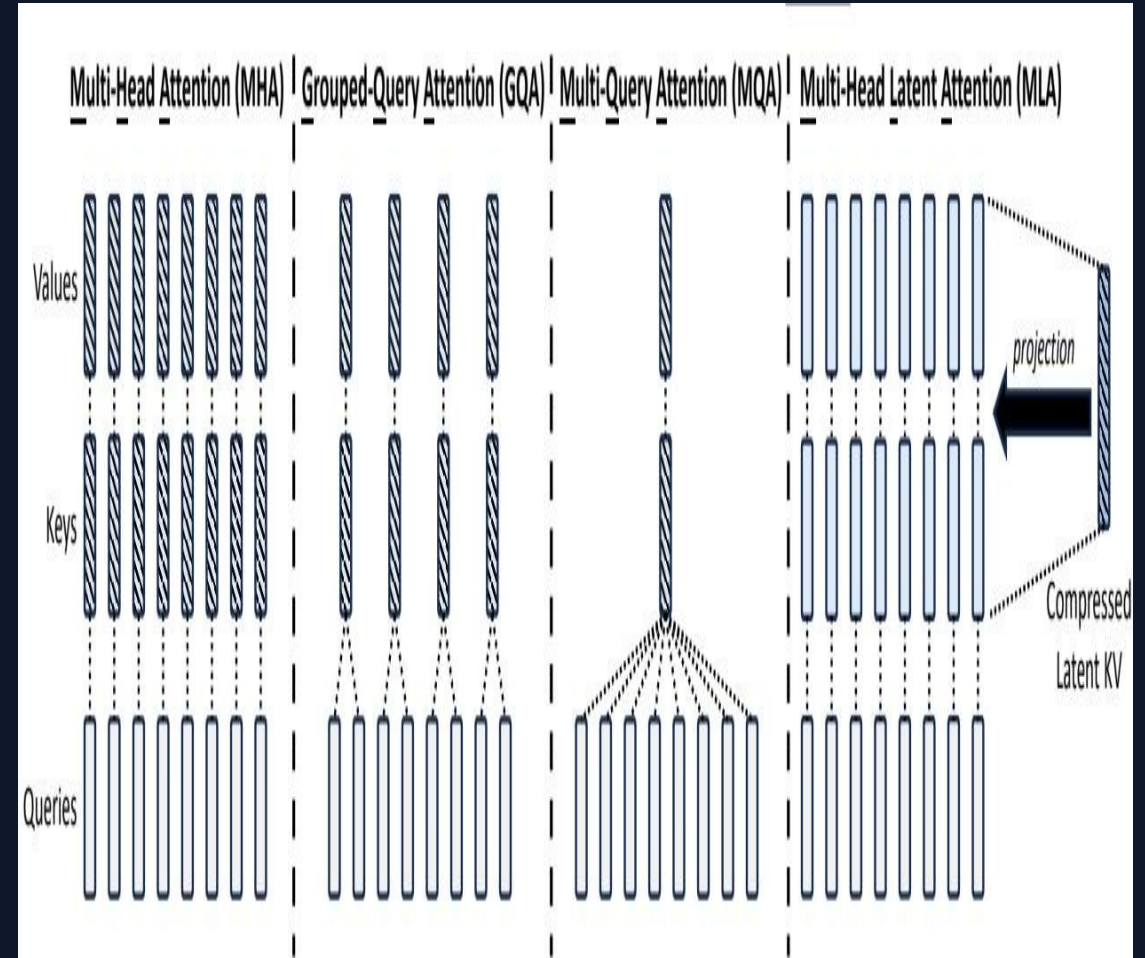
Decoupled RoPE

Positional encoding applied via a separate key component, preserving compatibility with RoPE while enabling the low-rank compression.

✓ Better performance than MHA

✓ 93.3% smaller KV cache

✓ Strictly superior to GQA/MQA



DeepSeekMoE Architecture

Fine-Grained Experts with Shared Expert Isolation

Expert Types

Shared Experts (N_s)

Always activated for every token.
Capture common, broadly useful knowledge.

Routed Experts (N_r)

Sparsely activated via top- K_r routing.
Specialize in distinct knowledge domains.

Fine-Grained Segmentation

Smaller, more numerous experts increase specialization and combinatorial flexibility.

Routing & Efficiency

Device-Limited Routing

Each token routed to experts on at most M devices.
Minimizes cross-node communication overhead.

Auxiliary Load-Balancing Losses

Enforce balanced utilization across experts and devices, preventing routing collapse.

Token-Dropping Strategy

Prevents expert overflow during training.
No token dropping at inference — no quality loss.

Result: Train 236B params at 42.5% less cost than a 67B dense model.

Efficiency Gains: DeepSeek-V2 vs. DeepSeek 67B

Training Cost, KV Cache, and Generation Throughput

DeepSeek 67B (Dense Baseline)

Architecture: Dense Transformer

Activated Params: 67B (all)

Training Cost

Baseline (1.0×)

KV Cache

Baseline (1.0×)

Generation Throughput

Baseline (1.0×)

DeepSeek-V2 (MoE + MLA)

Architecture: MoE + MLA

Activated Params: 21B of 236B

Training Cost

-42.5%

KV Cache Size

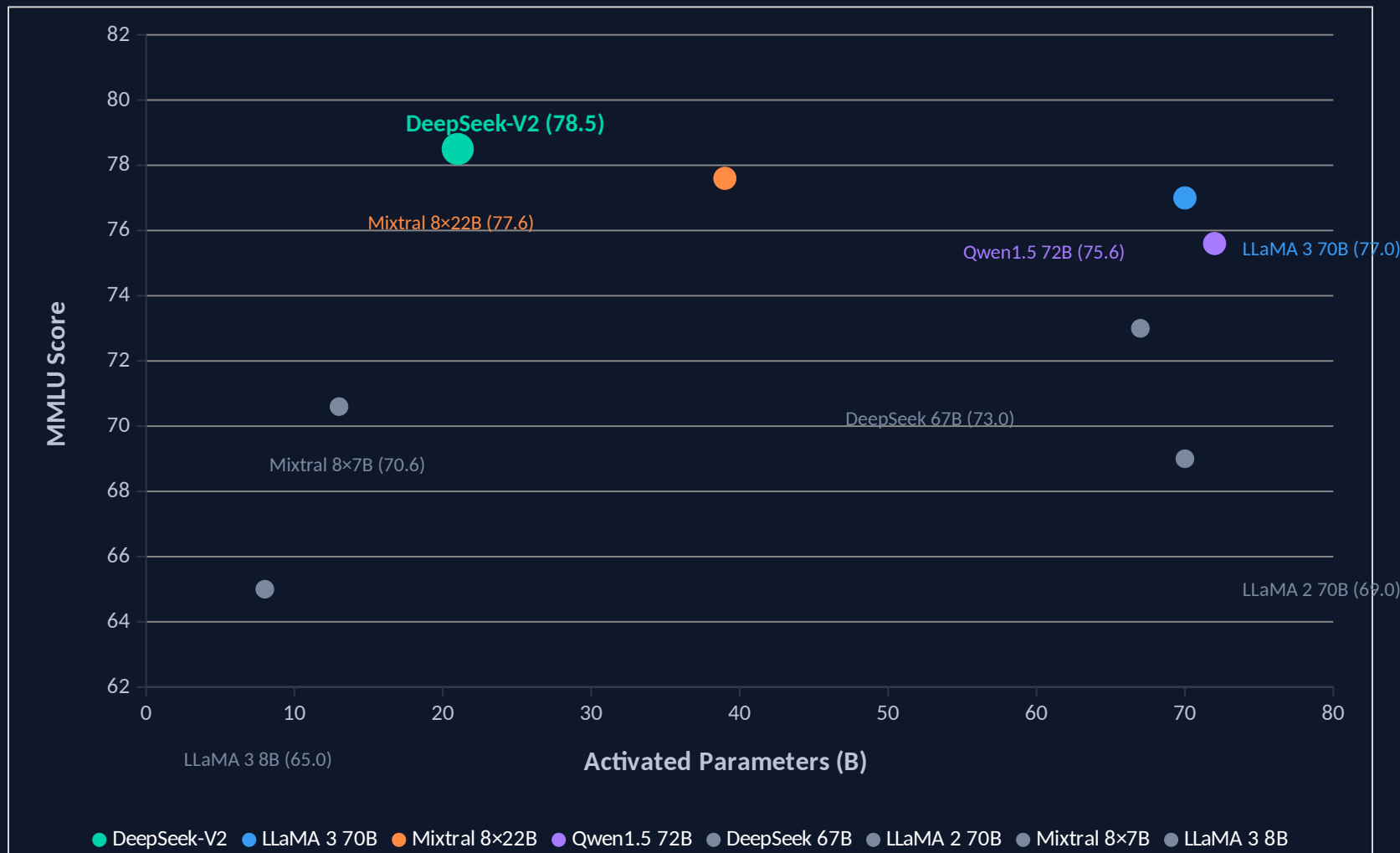
-93.3%

Generation Throughput

5.76× faster

MMLU vs. Activated Parameters

DeepSeek-V2 Achieves Pareto-Optimal Efficiency



Pareto-Optimal

DeepSeek-V2 achieves the highest MMLU score (78.5) among open-source models

while activating only 21B params — 3.3× fewer than LLaMA 3 70B.

Outperforms 70B-class dense models at a fraction of the compute.

Alignment & Chat Evaluation

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models

Alignment Pipeline

1. Supervised Fine-Tuning (SFT)

1.5M conversational sessions

2. Reinforcement Learning (GRPO)

Group Relative Policy Optimization

No critic model required

Chinese Language Leader

On AlignBench (Chinese), DeepSeek-V2 Chat (RL) outperforms:

- ✓ All open-source models
- ✓ Most closed-source models

AlpacaEval 2.0

38.9%

Length-controlled
win rate

MT-Bench

8.97

Overall score

AlignBench

7.91

Chinese overall
(#1 open-source)

Key Takeaways

Architectural Innovation Unlocks Efficient Scaling

MLA > MHA for Inference

Multi-Head Latent Attention achieves better performance than MHA while reducing KV cache by 93.3%. Strictly superior to GQA and MQA.

Economical Training via DeepSeekMoE

Fine-grained experts + shared expert isolation enable training a 236B-parameter model at 42.5% lower cost than a 67B dense model.

Pareto-Optimal Performance

21B activated parameters match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU and other benchmarks.

Deployment-Ready Throughput

5.76× generation throughput improvement over DeepSeek 67B makes large-scale deployment practical and cost-effective

Top-Tier Multilingual Alignment

Strongest open-source chat model in Chinese; competitive with closed-source systems in English

DeepSeek-V2 demonstrates that

**MoE + efficient attention = Pareto-optimal
LLM scaling**